EXPO
engineering ▲ design

TerraCrate

Secure, offline-first file storage for local networks

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Value Proposition

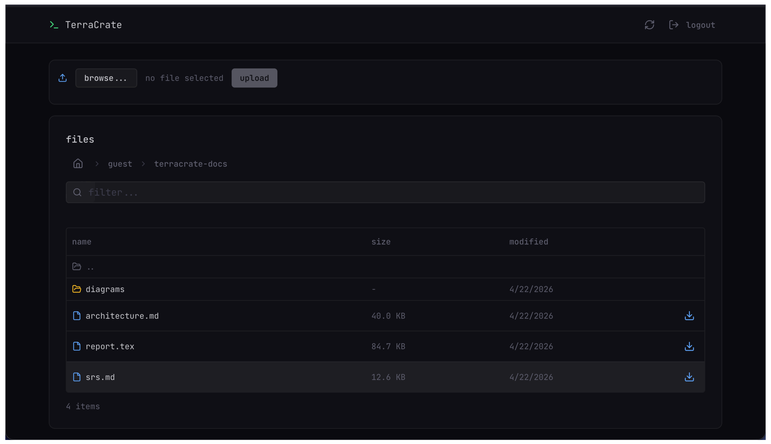
No lightweight solution combines certificate-based authentication, fine-grained access control, and zero-config discovery in one device.

Objective

Build a portable, self-hosted file server on a Raspberry Pi that provides secure multi-user file sharing with mTLS authentication, three-tier permissions, and mDNS discovery, deployable in any environment without cloud infrastructure or client software beyond a browser.

Problem Definition

Pocket-sized, self-contained file storage with certificate-based access. Plug in, sign up, and your certificate arrives by email. Unplug to take your data. No cloud, no residue.

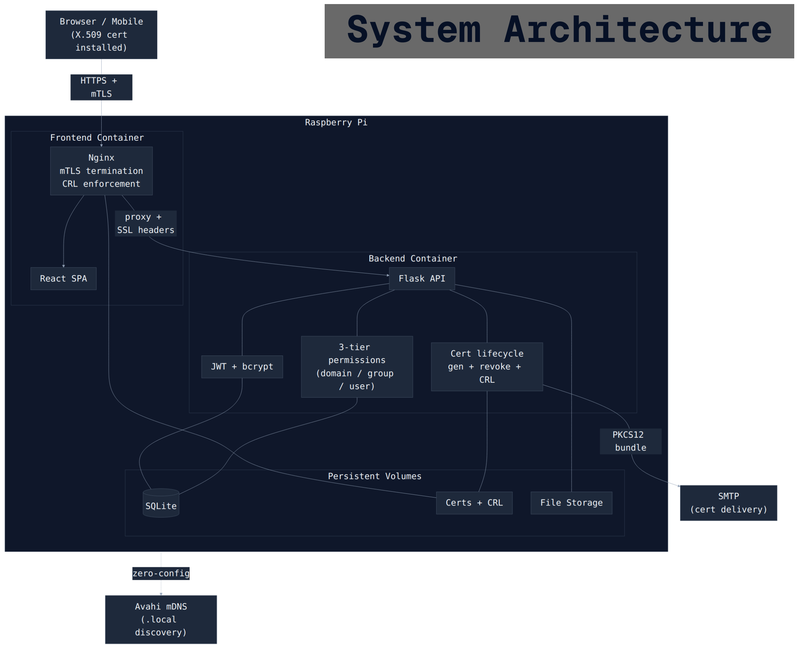


Users can browse shared files directly in the browser after installing their client certificate.

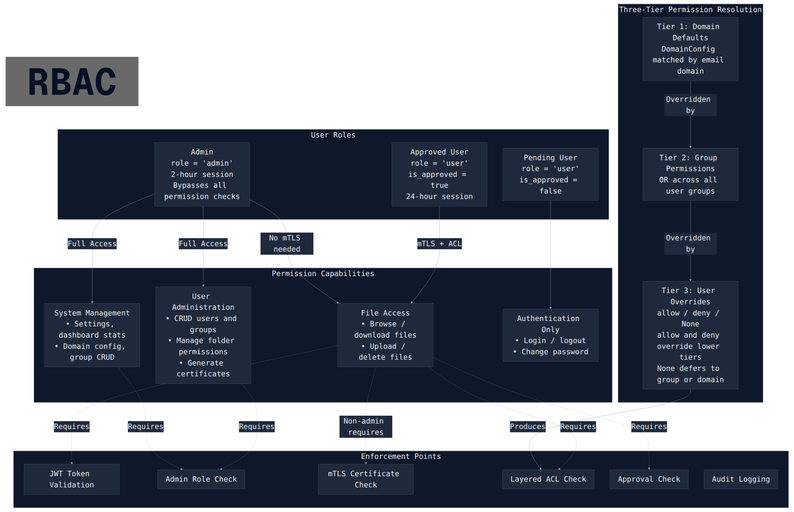
Key Requirements

All traffic uses HTTPS; regular users authenticate via mTLS. The full certificate lifecycle is managed on-device. A three-tier permission model covers domain, group, and per-user overrides. mDNS with Wi-Fi AP fallback enables zero-config discovery. All security events are logged and the system runs fully self-contained.

System Architecture



RBAC

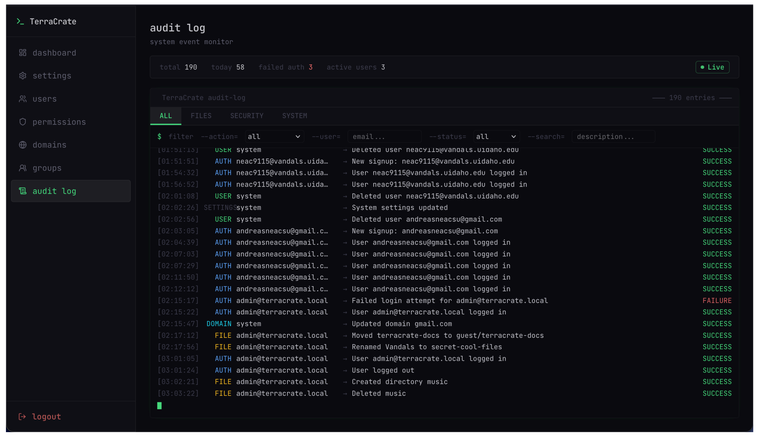


Concept Development

Two prior designs were evaluated: an SMB-over-mTLS approach requiring client-side Docker, and a QR code JWT approach lacking mutual TLS and per-user access control. The final design resolves both by delegating mTLS verification to Nginx, requiring only a browser with an installed certificate.

Validation

The backend is verified by 105 unit tests across authentication, permission resolution, certificate lifecycle, and path traversal prevention. The frontend is enforced through ESLint, TypeScript strict mode, and Vite builds. A three-workflow GitHub Actions pipeline runs linting, type checking, all backend tests, and Docker builds on every push.



The real-time audit log captures every security-relevant action with timestamp, user, and outcome for auditability.

Conclusion

TerraCrate delivers a secure, portable, self-hosted file sharing system with no cloud dependencies. mTLS verification delegated to Nginx provides cryptographic mutual authentication requiring only a browser. The three-tier permissions, certificate lifecycle, mDNS discovery, and audit logging satisfy all core requirements.

Acknowledgements

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